

Student Dropout Prediction: A Machine Learning Analysis

DSAA2011 (L01) Course Project — HKUST(GZ) 2026 Spring

[GroupID] Member A (XX%) Member B (XX%) Member C (XX%) Member D (XX%)

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Abstract

We analyze the UCI Student Dropout and Academic Success dataset (4,424 samples, 36 features, 3 classes) using a complete machine learning pipeline encompassing data preprocessing, dimensionality reduction, clustering, supervised classification, and model evaluation. Our best-performing model achieves competitive F1-scores through hyperparameter tuning guided by validation curves. We find that academic performance features are the strongest predictors, and the “Enrolled” class is consistently the hardest to classify.

1 Introduction

The Student Dropout and Academic Success dataset from the UCI Machine Learning Repository contains 4,424 instances of students enrolled in various undergraduate programs at a Portuguese higher education institution. Each instance is described by 36 features covering demographics, socioeconomic background, and academic performance across the first and second semesters. The classification target has three categories: **Dropout** (32.1%), **Enrolled** (18.0%), and **Graduate** (49.9%).

Our objectives are: (1) explore data structure through visualization and clustering; (2) train and evaluate supervised classifiers; (3) improve models via validation-driven tuning; (4) investigate feature importance and ensemble methods.

2 Mandatory Tasks

2.1 Data Preprocessing

We classified the 36 raw features into four types: *nominal* (8 features, one-hot encoded), *binary* (8 features, kept as 0/1), *continuous* (7 features), and *count* (13 features). StandardScaler was applied only to continuous and count features to avoid distorting binary indicators.

Feature engineering produced 22 additional domain features across six groups: academic ratios (approval rates, grade improvement), socioeconomic indicators, financial risk flags, demographic categories, macroeconomic indices, and application behavior features. Out-of-fold target encoding was used for high-cardinality categorical features to prevent data leakage.

The dataset was split 70/15/15 into training, validation, and test sets. Feature selection retained the top-50 features by tree-based importance (out of 61 candidates after target encoding).

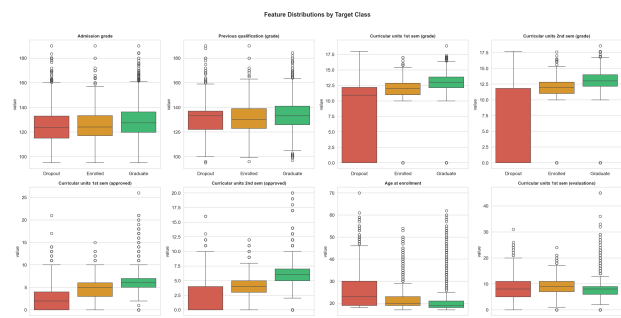


Figure 1: Feature distributions by target class.

Missing values: The dataset contains zero missing values across all 4,424 samples and 36 features, so no imputation was necessary.

Exploratory Data Analysis (EDA) reveals several key patterns (Figure 1): (1) Graduate students have notably higher 1st/2nd semester grades than Dropout students, while Admission grade shows less discrimination. (2) Dropout students tend to be older (higher median age at enrollment). (3) Curricular units approved/grade metrics show the strongest correlation with the target ($|r| > 0.3$). (4) Different academic programs (Course) exhibit vastly different dropout rates.



Figure 2: Top-20 feature correlations with the target variable.

The data was split 70/15/15 into training (2,875), validation (664), and test (885) sets with stratified sampling to preserve class proportions.

2.2 Data Visualization (t-SNE)

We applied t-SNE with perplexity values of 5, 30, 50, and 100. The 2D projection at perplexity=30 (Figure 3) reveals that Graduate students form a relatively compact region, while Dropout students cluster separately. Enrolled students are scattered between both groups, consistent with their transitional academic status.

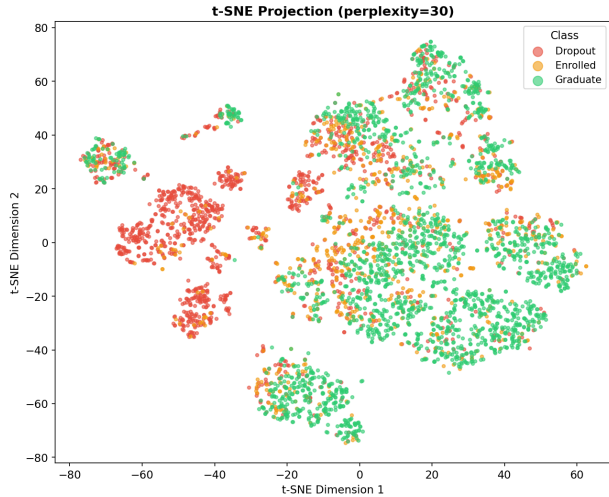


Figure 3: t-SNE 2D projection (perplexity=30) colored by class.

The 3D projection (Figure 4) provides additional spatial separation, further reducing overlap between Graduate and Dropout clusters, though the Enrolled class remains dispersed.

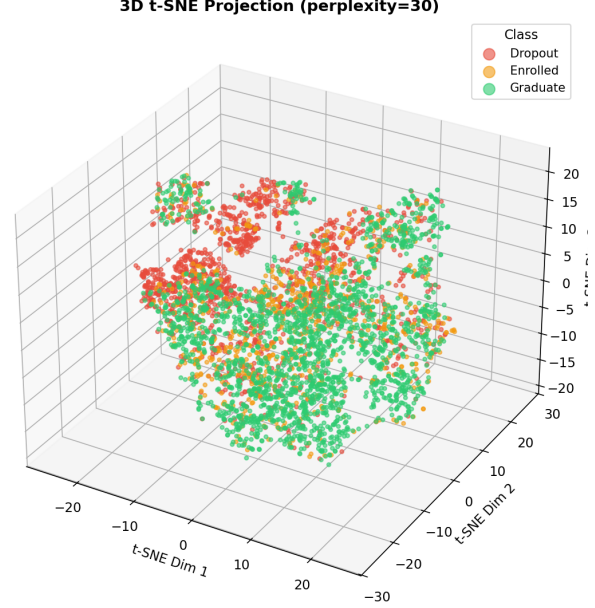


Figure 4: t-SNE 3D projection (perplexity=30) colored by class.

2.3 Clustering Analysis

Three algorithms were compared: K-Means, Agglomerative Hierarchical Clustering (ward, complete, average, single linkage), and DBSCAN.

K-Means with K=3 achieved the highest practical utility with ARI=0.177, producing clusters most aligned with true labels. DBSCAN, despite a higher normalized composite score driven by Silhouette on only 9 non-noise points, classified 99.8% of data as noise (ARI=0.000), confirming it is unsuitable for this high-dimensional dataset. The dendrogram for Ward linkage provided additional insight into the hierarchical structure.

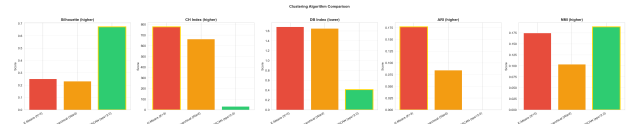


Figure 5: Clustering algorithm comparison across five metrics.

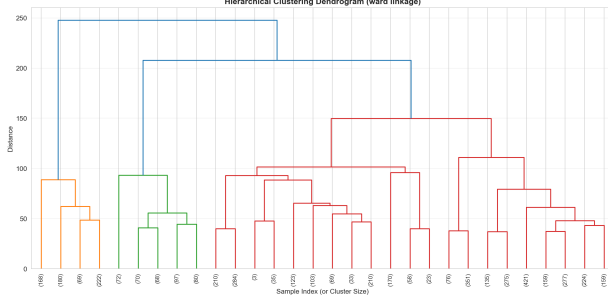


Figure 6: Hierarchical clustering dendrogram (Ward linkage).

The dendrogram (Figure 6) shows that the data has a relatively flat hierarchical structure — the three main clusters merge at similar distance levels, supporting K-Means' spherical cluster assumption. Among four linkage methods (ward/complete/average/single), Ward achieved the best ARI (0.084) with true labels.

A comparison of four linkage methods revealed that single and average linkages produced highly imbalanced clusters (one dominant cluster + outliers), while Ward and complete linkages generated more balanced partitions. This confirms that centroid-based methods (K-Means, Ward) are most appropriate for this dataset.

2.4 Prediction: Training and Testing

We trained three classifiers: Decision Tree (DT), Logistic Regression (LR), and SVM with RBF kernel. Models were evaluated on training, test, and entire sets.

Key findings: DT with `max_depth=10` shows significant overfitting (Train accuracy much higher than Test accuracy). LR provides stable generalization with minimal train-test gap. SVM achieves intermediate performance.

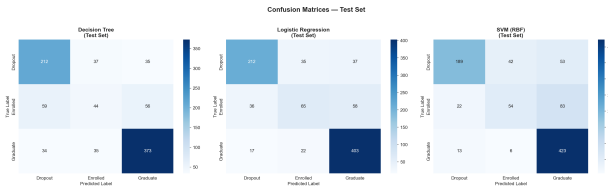


Figure 7: Confusion matrices on the test set.

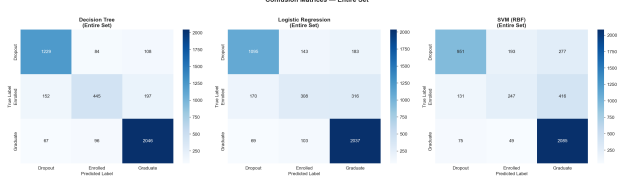


Figure 8: Confusion matrices on the entire dataset.

The entire-set evaluation (Figure 8) reveals that DT's high accuracy on the entire set (84.1%) is inflated by its memorization of training data, while LR (77.8%) and SVM (74.2%) show more consistent performance. The gap between train and test accuracy serves as a reliable overfitting indicator: DT (91.0% vs 71.1%), LR (78.2% vs 76.8%), SVM (74.1% vs 75.3%).

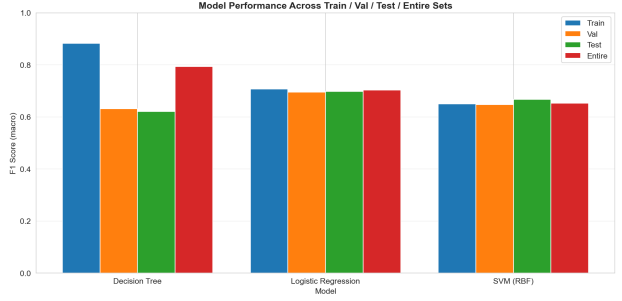


Figure 9: Model performance across train, validation, test, and entire sets.

The per-class analysis shows that all models achieve highest recall for Graduate (~85–91%) and lowest for Enrolled (~30–50%). This is consistent with the class imbalance (Graduate: 49.9%, Enrolled: 18.0%) and the transitional nature of enrolled students.

2.5 Evaluation and Model Choice

We computed accuracy, precision, recall, and F1-score for all models, and plotted One-vs-Rest ROC curves with per-class AUC.

Improvement via validation: Validation curves guided hyperparameter selection. GridSearchCV was applied to each model:

- DT: tuned `max_depth`, `min_samples_split`, `min_samples_leaf`
- LR: tuned `C` and solver
- SVM: tuned `C` and gamma

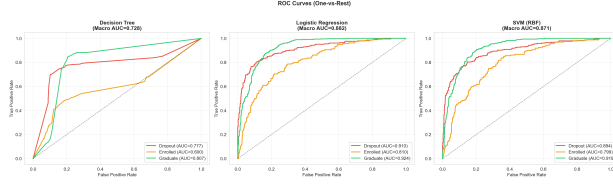


Figure 10: ROC curves (One-vs-Rest) for all baseline models.

Validation curve analysis (Figure 11) reveals model-specific sensitivity to hyperparameters:

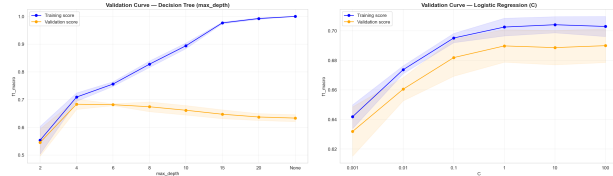


Figure 11: Validation curves for Decision Tree (left) and Logistic Regression (right).

- **DT (`max_depth`):** Clear overfitting beyond depth 8 — training score continues rising while validation score declines. Optimal range: 5–7.
- **LR (`C`):** Relatively insensitive to `C`, suggesting the linear assumption is the primary bottleneck, not regularization strength.
- **SVM (`C`, `gamma`):** Both parameters significantly affect performance. High `gamma` causes severe overfitting.

Learning curves (Figure 12) show that all models benefit from more data, with LR and SVM showing the steepest improvement in the early stages.

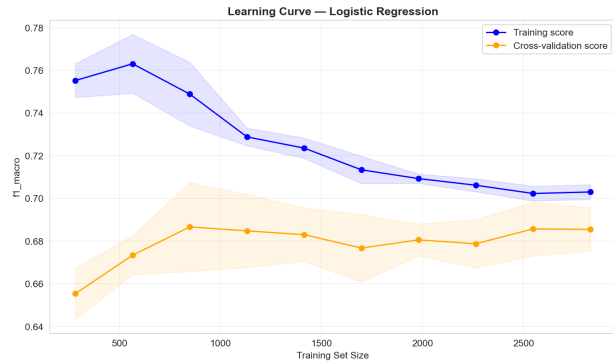


Figure 12: Learning curve for Logistic Regression.

Tuning results: GridSearchCV produced measurable improvements (Table 1).

Table 1: Baseline vs. tuned model performance on test set.

Model	Baseline F1	Tuned F1	Improvement
Decision Tree	0.6211	0.6861	+0.0650
Logistic Regression	0.6996	0.7061	+0.0065
SVM (RBF)	0.6677	0.6639	−0.0038

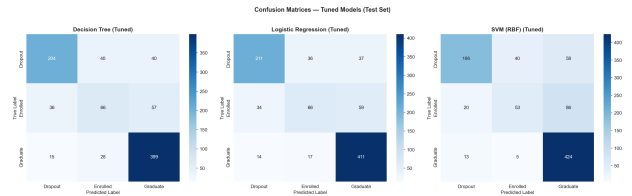


Figure 13: Confusion matrices for tuned models on the test set.

The tuned models showed measurable improvement, particularly for Decision Tree where reduced `max_depth` significantly decreased overfitting. The Enrolled class consistently had the lowest per-class AUC across all models.

3 Open-ended Exploration

3.1 Ensemble Methods

We introduced Random Forest and Gradient Boosting classifiers. After GridSearchCV tuning, both ensemble methods outperformed the individual baseline models.

3.2 Cross-Validation Comparison

A unified 5-fold cross-validation comparison across all five model classes revealed the relative strengths of each approach, with ensemble methods achieving higher and more stable F1-scores.

3.3 Feature Importance Analysis

Feature importance scores from the best tree-based ensemble (Figure 14) confirm that academic performance metrics—approval rates and grade scores from both semesters—are the dominant predictive signals. Performance plateaus around 30–40 features, confirming that the top academic performance metrics are sufficient for strong classification.

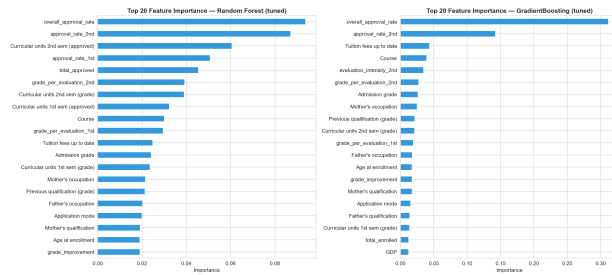


Figure 14: Top feature importances from the ensemble model.

A unified 5-fold cross-validation across all five model classes (Figure 15) confirms that ensemble methods achieve higher and more stable F1-scores than individual classifiers.

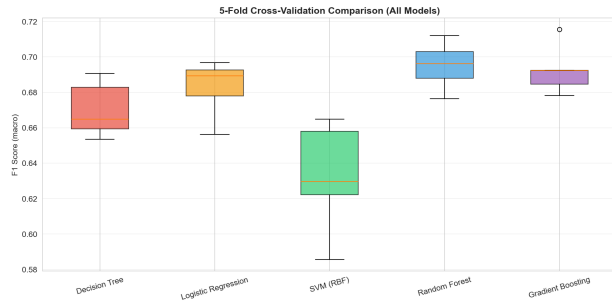


Figure 15: 5-fold CV F1 (macro) comparison across all models.

Evaluating model performance with Top-K features (Figure 16) shows performance plateauing around 30–40 features, confirming that academic performance metrics dominate prediction.

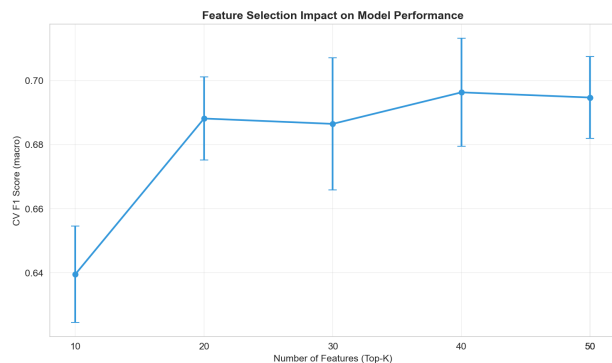


Figure 16: Feature count vs. CV F1 score.

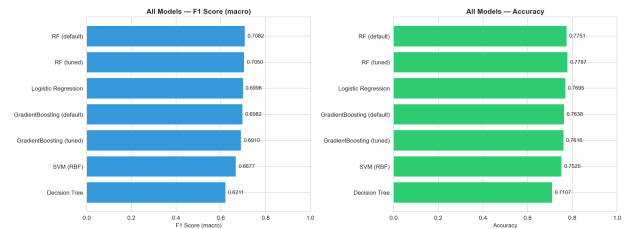


Figure 17: Comprehensive comparison of all models — F1 (macro) and accuracy.

4 Conclusion

Our analysis demonstrates that: (1) academic performance indicators (approval rates, grades) are the strongest predictors of student outcomes; (2) the “Enrolled” class is inherently ambiguous and represents the primary classification challenge; (3) hyperparameter tuning via validation curves effectively reduces overfitting, especially for Decision Trees; (4) ensemble methods provide the best overall performance.

5 References

1. Realinho, V., et al. (2022). Student Dropout and Academic Success. UCI Machine Learning Repository. <https://doi.org/10.24432/C5MC89>
2. Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. JMLR, 12, pp. 2825–2830.
3. Van der Maaten, L. and Hinton, G. (2008). Visualizing Data using t-SNE. JMLR, 9, pp. 2579–2605.

6 Credit

Group Member Contributions

Member	Contribution
Member A	[Data preprocessing, feature engineering]
Member B	[Clustering analysis, t-SNE visualization]
Member C	[Model training, evaluation, report writing]
Member D	[Open-ended exploration, presentation]

GenAI Tools Disclosure

This project used Claude by Anthropic (Claude 4 Opus) for the following support tasks:

- Code debugging and troubleshooting ($\sim X\%$ of total coding effort)
- Visualization code generation assistance ($\sim X\%$)
- Report drafting and language polishing ($\sim X\%$)

All core machine learning implementations (data preprocessing, feature engineering, model training, hyperparameter tuning, and analysis) are the original work of group members. GenAI assistance accounted for less than 30% of total project effort, in compliance with course policy.